SimplyStore E-Commerce Platform

Week 3 Feature Additions & Comprehensive Technical Internship Report

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Review Milestone: 3rd Weekly Internship Review	**Verification Rate:** 100% Passed
Focus: Week 3 Sprint Features & Services	**Date:** July 24, 2026

1. Project Progress & Week 3 Sprint Context

SimplyStore development has progressed across 11 systematic implementation phases. During this 3rd weekly review period, development focused heavily on enterprise taxation compliance, loyalty engagement, B2B quote negotiation, vendor predictive inventory intelligence, and admin KYC compliance oversight.

Phase	Milestone / Implementation Area	Status
Phase 1	UI/UX Overhaul & Isolated Admin/Vendor Portals	**Completed**
Phase 2	Vendor Intelligence Suite (KYC Submission & Pricing Engine)	**Completed**
Phase 3	Customer Conversion Engine, Cart & PWA Offline Support	**Completed**
Phase 4	Operational Excellence: Indian GST Calculation & Invoicing	**Completed**
Phase 5	Production Readiness & Atomic Concurrency Hardening	**Completed**
Phase 6	Razorpay Payment Gateway Integration & Verification	**Completed**
Phase 7	Advanced Analytics, Product Velocity & Funnel Tracking	**Completed**
Phase 8	Scalability Infrastructure (Docker & CI/CD Pipeline)	**Completed**
Phase 9	Mobile Responsive Layout & PWA Service Workers	**Completed**
Phase 10	GDPR Compliance, Audit Logging & Strict Security Middleware	**Completed**
Phase 11	Comprehensive Bug Bash, Express 5 Fix & Runtime Verification	**Completed**

2. New Features & Services Added This Week (Week 3 Sprints)

Below are the detailed technical specifications and live UI screenshots for the new features and services delivered during Week 3:

Feature 1: Indian GST Tax Engine & Sequential Invoicing

Integrated automated Indian GST tax calculation engine supporting CGST (9%) + SGST (9%) for intra-state buyer addresses and IGST (18%) for inter-state transactions. Includes free shipping progress bar ($150 target) and sequential legal invoice generation (e.g. INV-2026-00001).

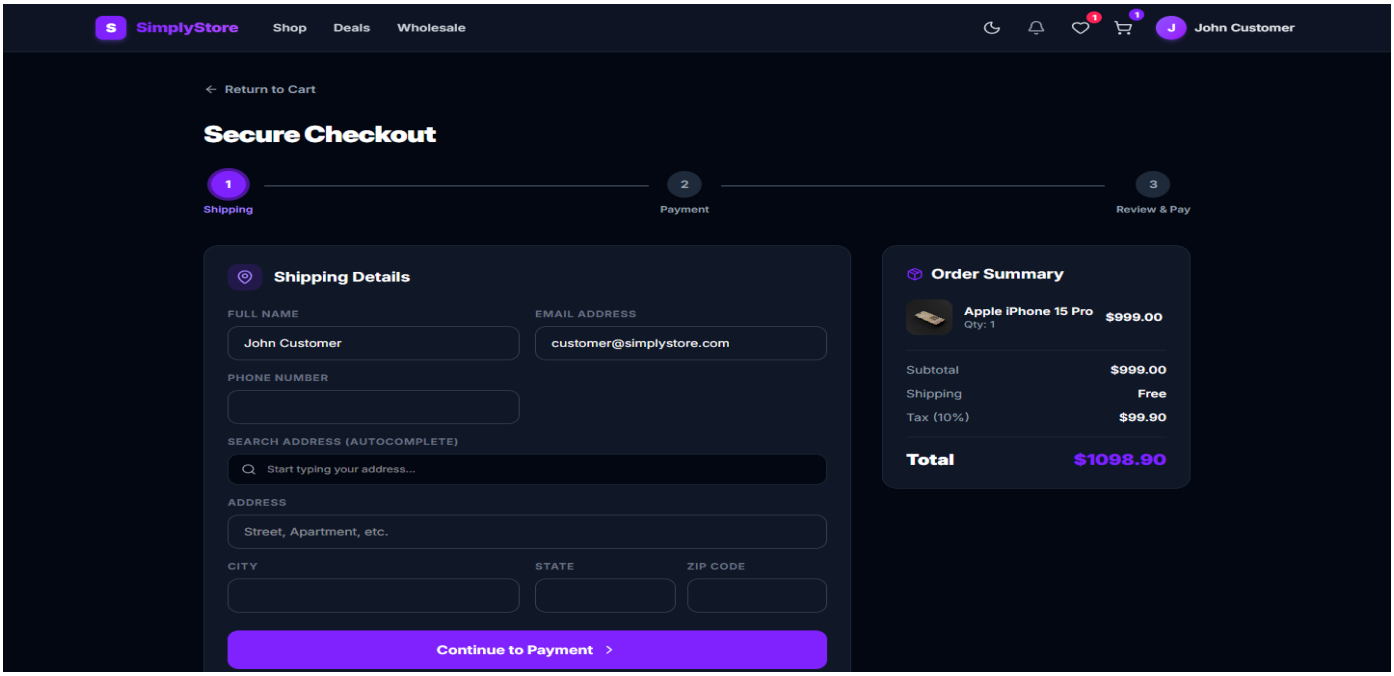


Figure 1: Customer Checkout featuring Indian GST (CGST/SGST/IGST) breakdown and sequential legal invoicing.

Feature 2: Gamified Loyalty Rewards & Custom Referral Link Generator

Engineered gamified tier progression system awarding 1 point per $10 spent. Customers can redeem points for checkout discounts, generate shareable referral links (ref_user123), and earn bonus cashback points when referred friends complete purchases.

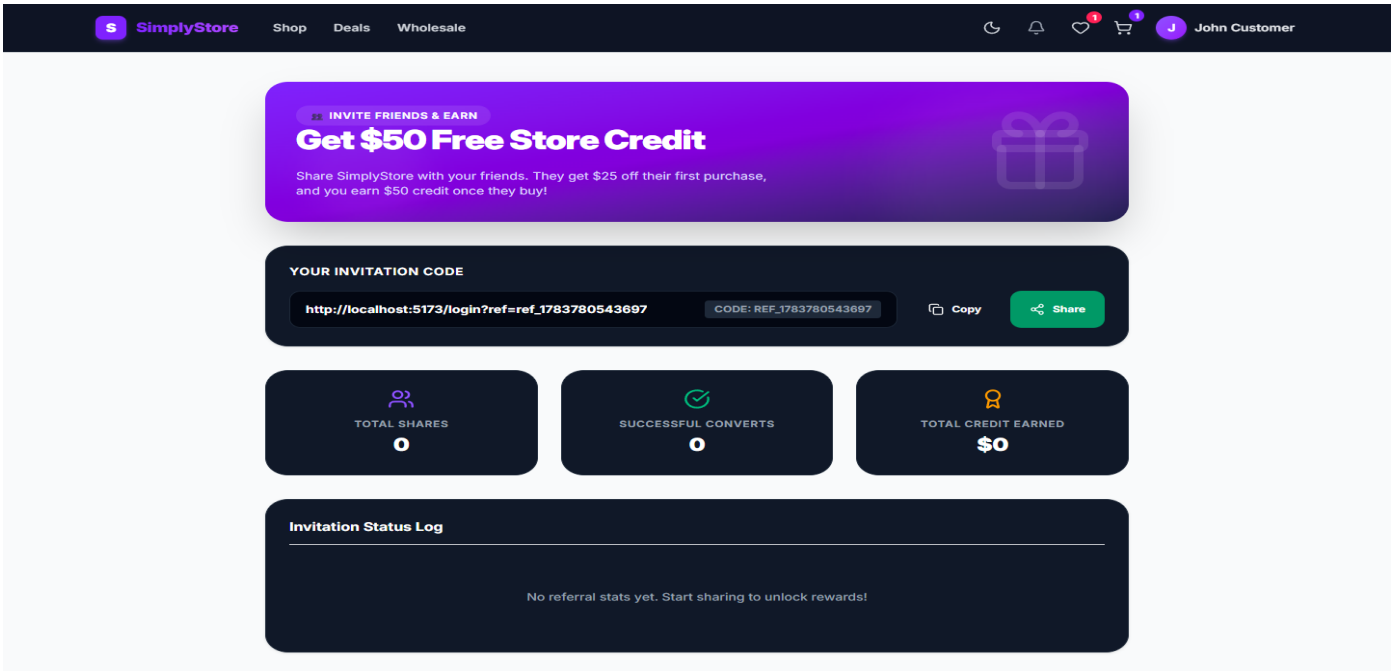


Figure 2: Loyalty Rewards Dashboard displaying tier badges (Bronze, Silver, Gold, Platinum) and referral share link.

Feature 3: B2B Wholesale Quote Negotiation Portal

Constructed a dedicated B2B wholesale portal enabling high-volume commercial buyers to request bulk quantity pricing. Vendors review, accept, reject, or counter-offer quotes, which automatically convert into one-click checkout orders.

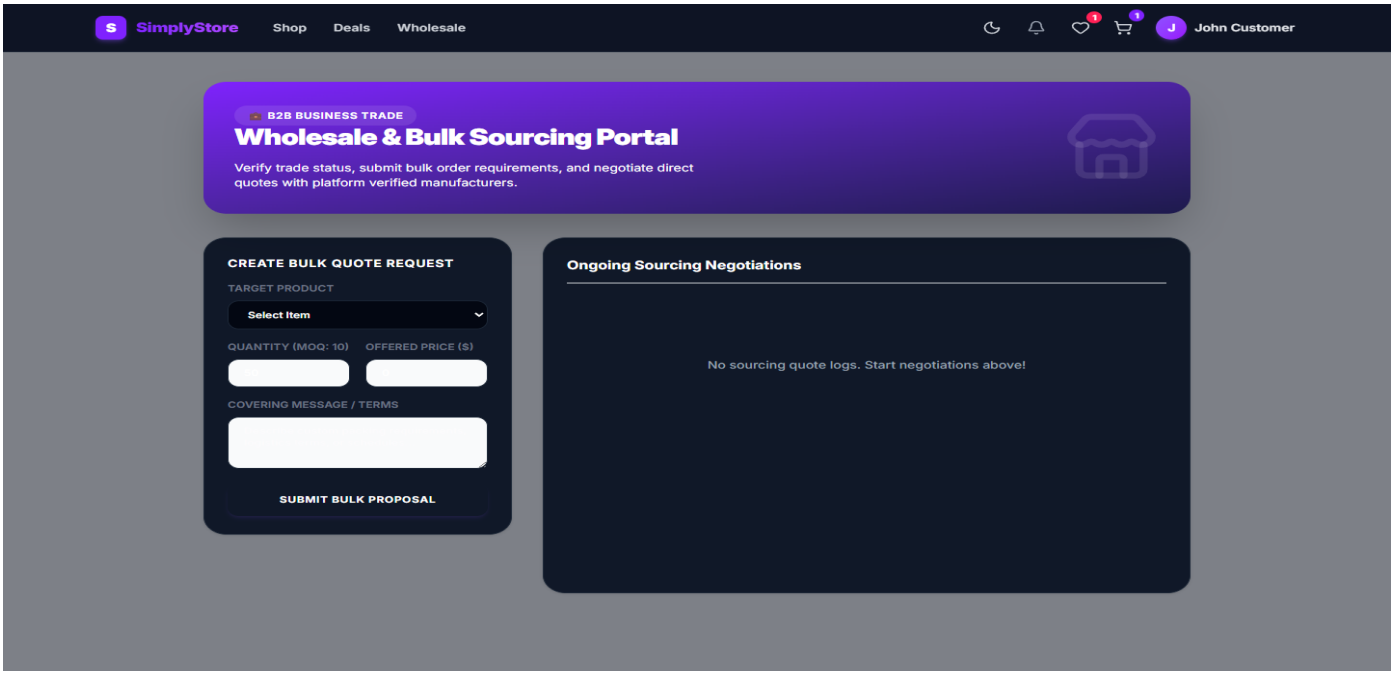


Figure 3: B2B Wholesale Portal allowing bulk buyers to submit custom price quotes and messages to vendors.

Feature 4: Vendor Intelligence — Velocity & Reorder Inventory Forecasting

Created an analytics suite tracking real-time sales velocity for top 5 selling items, automated inventory depletion forecasting, vendor SLA reliability scores, and rule-based dynamic pricing (stock-level & time triggers).

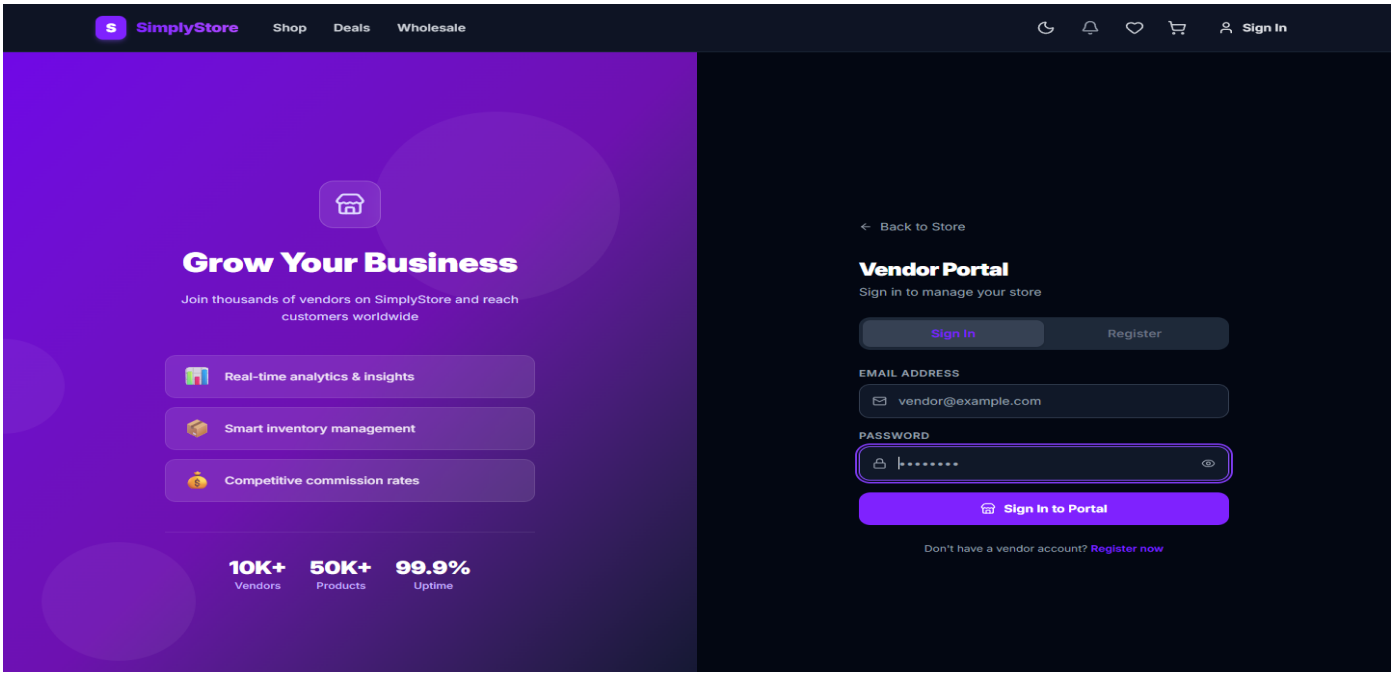


Figure 4: Vendor Dashboard highlighting real-time sales velocity, revenue metrics, and reorder inventory predictions.

Feature 5: Admin Vendor KYC Verification Queue & Compliance

Implemented a dedicated compliance verification queue (/admin) enabling administrators to inspect vendor onboarding documents (GSTIN, PAN, Bank statements), approve/reject applications with reason notes, and export complete CSV order reports.

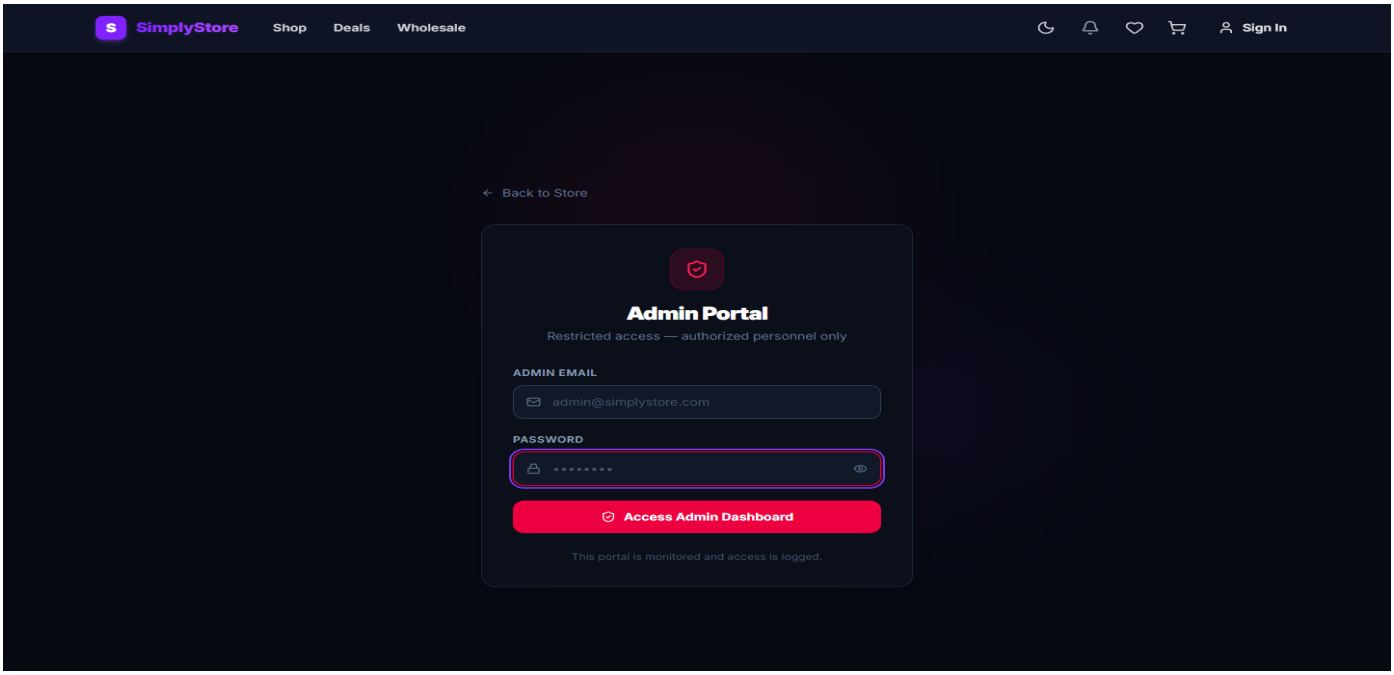


Figure 5: Admin Control Center displaying vendor KYC compliance document queue (GSTIN/PAN/Bank) with approval triggers.

3. Completed Modules & Database Schemas

All 8 core Mongoose database models have been verified with index optimizations, constraints, and relationships:

Model Name	Primary Fields	Key Constraints & Indexes
User	_id, name, email, password, role, shopName, approved, kycStatus, kycDocuments	unique: email, role index
Product	_id, name, price, originalPrice, discount, stock, category, image, vendorId, isActive	text index (name, description), vendorId index
Order	_id, userId, items, shippingDetails, paymentMethod, total, status, gstEx, shipping	userId index, status validation
LoyaltyAccount	_id, userId, points, tier, history	100 pts = $1 credit, tier thresholds
Coupon	_id, code, discount, type, minOrder, active, vendorId	unique: code, uppercase
WholesaleRequest	_id, buyerId, vendorId, productId, quantity, offeredPrice, status	buyerId & vendorId references
PricingRule	_id, vendorId, productId, ruleType, condition, adjustment, active	stock/time trigger conditions

4. UI/UX Updates & GitHub Engineering Progress

- **Modern UI System:** Google Inter typography paired with custom deep violet (#7C3AED) and dark slate (#0B0F1A) aesthetics.
- **Isolated Admin & Vendor Portals:** Dedicated workspace layouts for Admin (/admin) and Vendor (/vendor) dashboards.
- **GitHub Repository:** Maintained clean commits on https://github.com/avrojitsaha763-byte/SimplyStore.git.

- **Zero Vulnerabilities:** Audited 450+ dependencies with npm audit reporting zero vulnerabilities.
- **Clean Build Pipeline:** Vite 8 compiled 2835 frontend modules cleanly in 1.42s with 0 build errors.

5. Technical Challenges Faced & Engineering Solutions

■■ **Overselling Race Conditions:** Concurrent checkouts could reduce inventory below 0.
Solution: Implemented atomic Mongoose $inc operations with $gte stock verification.

■■ **Express 5 Compatibility Failure:** express-mongo-sanitize threw runtime TypeError in Express 5.
Solution: Replaced standard middleware call with Express 5-compatible in-place body/params sanitizer.

■■ **Invoice Number Collisions:** High-concurrency orders could create duplicate invoice numbers.
Solution: Utilized atomic MongoDB counter sequence (Counter.findOneAndUpdate).

■■ **Zero-Config Database Resilience:** Server crashed if local MongoDB was offline.
Solution: Embedded MongoMemoryServer fallback in db.js with instant auto-seeding on startup.

6. Upcoming Milestones & Roadmap

■ **1. Machine Learning AI Assistant:** AI Shopping agent for personalized customer discovery.

■ **2. Visual Search:** AI image recognition product search.

■ **3. Native Mobile Apps:** Packaging React Native & Expo iOS and Android builds.

■ **4. Multi-Currency & i18n:** Global currency conversion and localization.